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These are the most asked interview questions compiled by Maitreya Ranade for the role of FPGA engineer.

1 Table of Contents

1.1	Digital Design
1.2	FPGA Architecture
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1.6	Writing HDL Codes

Sort:

What is FIFO? Explain Synchronous vs Asynchronous FIFO. Explain Dual port RAM? What are full & empty conditions in FIFO? How to get 50% Duty Cycle for the same How to get 25% and 75% Duty Cycle for the same

1.7 # Digital Design Interview Questions

- Digital Design Interview Questions
 - Combinational logic design
 - Sequential logic design
 - Finite state machines (FSMs)
 - Sort
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2 Combinational logic design

1. What is a combinational circuit?
2. Difference between combinational & sequential circuits
3. What are universal gates?
4. Which logic gate is used more in real-life applications?
5. What are the applications of Buffer?
6. Realize the equation $Y=AB+C$ using only 2 input NAND gates
7. Realize XOR gate using only 2 input NOR gates
8. Implement a NOT gate using the XOR gate
9. Explain the difference between NAND and negative input AND gate
10. Explain the differences between MOSFETs and BJTs, and why we use MOSFET in digital design.
11. Draw 2 input NAND & NOR gates using CMOS logic
12. What are pull-up & pull-down networks?
13. Explain the operation of the CMOS Inverter
14. Draw the VTC curve of the CMOS Inverter
15. Explain about half adder
16. Design an OR gate using half adders
17. Design a Full adder using only NAND gates
18. In how many ways, a full adder can be implemented using logic gates?
19. Draw the Full subtractor truth table & circuit
20. Difference between Ripple Carry Adder and Carry Look Ahead Adder
21. Design a logical circuit that performs both addition and subtraction
22. What is a comparator?
23. Differences between Encoder and Decoder
24. Applications of Encoder & Decoder
25. Design 4 to 16 decoder using 2 to 4 decoders
26. What is a priority encoder?
27. What is a Multiplexer? Applications of MUX
28. Why MUX is called a universal circuit?
29. Design 4:1 MUX using 2:1 MUX
30. Design logic gates (AND, OR, NOT, NAND, NOR, XOR) using 2:1 MUX
31. How many 2:1 MUX are required to design 64:1 MUX?

32. Realize 2 input AND gate using 4:1 MUX
33. Implement 2 input NOR gate using 1:2 DEMUX
34. Implement a full adder using 4:1 Muxes
35. Explain tri-state buffers (notif1, notif0, bufif1, bufif0)
36. Implement basic gates (AND, OR, NOT) using tri-state buffers
37. Design a 4:1 MUX using tri-state buffers
38. Design a 2:1 MUX using decoder & tri-state buffers
39. Design a combinational circuit that converts a 4-bit gray code number to a 4-bit binary number using only XOR gates
40. Design a combinational circuit, that doubles the input frequency. (frequency multiplier by 2)

Topics to revise: * Adders and subtractors * Multiplexers and demultiplexers * Encoders and decoders * Comparators * Parity generators and checkers * Barrel shifters * Arithmetic logic units (ALUs) * Half and full adders * Half and full subtractors

3 Sequential logic design

1. What is a latch? Explain using NOT, NAND, and NOR gates.
2. What is a latch? Explain using NOT, NAND, and NOR gates.
3. Why do we use NAND gates more when compared with NOR gates?
4. Difference between Latch and Flip-flop?
5. Why flip-flops are preferred when compared with latches?
6. What is a transparent latch?
7. What is race condition?
8. What is master-slave configuration?
9. What is a glitch?
10. Which is more sensitive to glitches (latch or flip-flop)?
11. Draw & explain D-flip flop vs D-latch timing diagrams.?
12. Draw a D-latch using 2:1 MUX.?
13. Draw a D-flip flop using 2:1 MUX.?
14. Which flip flop is most widely used?
15. Application of flip flops?
16. Draw the excitation table for the SR flip-flop.?
17. Design a JK flip-flop using a D flip-flop?
18. Design a T flip-flop using a D flip-flop?
19. Design a 4-bit up-down counter?
20. Design a frequency divider by 3 circuit.?
21. Frequency Multiplier Questions
22. Frequency Divider Questions (Understand all Frequency Division,duty cycle and waveforms)
23. Design & explain the MOD-5 counter.?
24. How many flip-flops are required to design a MOD-535 counter?
25. Draw a 4-bit counter using one D flip-flop & one adder circuit.?
26. Difference between synchronous and asynchronous counters?
27. What is asynchronous reset & how is it different from synchronous reset?
28. How many bits a flip-flop can store?

29. What is a register & how many bits it can store?
 30. What is a shift register & the types of shift registers?
 31. Draw a 4-bit Johnson counter?
 32. Design a PISO Shift register?
 33. Draw & Explain the 4-bit Linear Feedback Shift Register.?
 34. Draw & Explain a 4-bit Barrel Shifter using MUX.?
 35. Draw a state diagram for the 101 sequences? (Both Mealy & Moore non-overlap)
 36. Draw a state diagram for the JK flip-flop.?
 37. Design a sequence detector that detects both 1010 & 0101(both mealy & moore overlap).?
 38. Design a sequence detector that detects 1X1X (moore overlap).?
 39. Draw Block Diagram of D FlipFlop and Draw Waveform for the drawn D FlipFlop
 40. Draw a circuit for getting $0 \rightarrow 2 \rightarrow 4$ and repeat the sequence using any FlipFlop
 41. What is Clock Gating ? Explain How it is useful and how it works .
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4 Finite state machines (FSMs)

1. What is a Finite State Machine?
 2. Explain Mealy Vs Moore FSM.?
 3. Which is preferred (Mealy or Moore)?
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5 Sort

6 STA And CDC

1. What is Set up Time ?
2. What is Hold Time ?
3. How to Eliminate Set up time and Hold Time Violation ?
4. When operating frequency is 100MHz, there is Setup time violation and Hold time Violation. Then We are reducing frequency to 50MHz , then What will happen to Setup time violation and Hold time violation ?(Please Understand STA Very Well - Very Important)
5. What is a Clock Domain Crossing (CDC)?
6. How to overcome CDC?
7. What is a 2-stage synchronizer?
8. What is MTBF?
9. Explain Clock to Q (C2Q) delay.?
10. What is the setup time & hold time?
11. What is metastability?
12. How to avoid setup time & hold time violations?
13. What is clock skew?

7 HDL Theory questions

- HDL Theory questions
- Verilog

- SystemVerilog
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8 Verilog

1. Why do we need an HDL?
 2. Difference between Programming Language & Hardware Description Language
 3. What are the different types of HDLs present in the industry?
 4. Explain VHDL vs Verilog HDL
 5. Different levels of Abstraction in Verilog
 6. Explain the data types in Verilog
 7. Difference between reg and wire & default values of reg and wire
 8. Declare a 2-dimensional unpacked array
 9. Difference between \$display, \$strobe, \$write and \$monitor
 10. Difference between \$finish, \$stop and \$reset
 11. Write a Verilog code for NAND Gate using switch-level modelling
 12. What are the different gate primitives in Verilog? Explain user-defined primitives
 13. Explain the delays in Gate Level Modelling
 14. What are the different types of operators in Verilog?
 15. Difference between == and ===
 16. Explain logical and arithmetic shifts with examples
 17. Write a Verilog code for the decoder using conditional & shift operator
 18. Explain the delays in Data Flow Modelling
 19. What are procedural blocks in Verilog and explain them
 20. Difference between the initial block and the final block
 21. Explain the difference between 'begin-end' and 'fork-join'
 22. Blocking vs Non-blocking assignments
 23. Explain inter and intra-assignment delays with an example
 24. What is a parameter?
 25. Explain compiler directives
 26. Explain 'timescale in Verilog
 27. Explain Case-x and Case-z statements
 28. Explain the full case and parallel case statements
 29. Difference between task and function
 30. Explain the delays in Behavioural Modelling
 31. Explain simulation & synthesis
 32. What are the synthesizable constructs in Verilog?
 33. What is linting?

 34. Explain Event regions in Verilog
 35. What is structural modelling?
 36. What is meant by a generic style of coding?
 37. Explain port mapping in Verilog.
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9 SystemVerilog

1. What is the difference between an initial and final block of the systemverilog?
2. Explain the simulation phases of SystemVerilog verification?
3. What is the Difference between SystemVerilog packed and unpacked array?
4. What is “This ” keyword in the systemverilog?
5. What is alias in SystemVerilog?
6. randomized in the systemverilog test bench?
7. in SystemVerilog which array type is preferred for memory declaration and why?
8. How to avoid race round condition between DUT and test bench in SystemVerilog verification?
9. What are the advantages of the systemverilog program block?
10. What is the difference between logic and bit in SystemVerilog?
11. What is the difference between datatype logic and wire?
12. What is a virtual interface?
13. What is an abstract class?
14. What is the difference between \$random and \$urandom?
15. What is the expect statements in assertions?
16. What is DPI?
17. What is the difference between == and === ?
18. What are the system tasks?
19. What is SystemVerilog assertion binding and advantages of it?
20. What are parameterized classes?
21. How to generate array without randomization?
22. What is the difference between always_comb() and always@(*)?
23. What is the difference between overriding and overloading?
24. Explain the difference between deep copy and shallow copy?
25. What is interface and advantages over the normal way?
26. What is modport and explain the usage of it?
27. What is a clocking block?
28. What is the difference between the clocking block and modport?
29. System Verilog Interview Questions, Below are the most frequently asked questions.
30. What are the different types of verification approaches?
31. What are the basic testbench components?
32. What are the different layers of layered architecture?
33. What is the difference between a \$rose and @ (posedge)?
34. What is the use of extern?
35. What is scope randomization?
36. What is the difference between blocking and non-blocking assignments?
37. What are automatic variables?
38. What is the scope of local and private variables?
39. How to check if any bit of the expression is X or Z?
40. What is the Difference between param and typedef?
41. What is ‘timescale?
42. Explain the difference between new() and new[] ?
43. What is the difference between task and function in class and Module?
44. Why always blocks are not allowed in the program block?
45. Why forever is used instead of always in program block?
46. What is SVA?

47. Explain the difference between fork-join, fork-join_none, and fork-join_any?
48. What is the difference between mailboxes and queues?
49. What is casting?
50. What is inheritance and polymorphism?
51. What is callback?
52. What is constraint solve-before?
53. What is coverage and what are different types?
54. What is the importance of coverage in SystemVerilog verification?
55. When you will say that verification is completed?
56. What are illegal bins? Is it good to use it and why?
57. What is the advantage of seed in randomization?
58. What is circular dependency?
59. What is “super”?
60. What is the input skew and output skew in the clocking block?
61. What is a static variable?
62. What is a package?
63. What is the difference between bit [7:0] and byte?
64. What is randomization and what can be
65. What are the constraints? Is all constraints are bidirectional?
66. What are in line constraints?
67. What is the difference between rand and randc?
68. Explain pass by value and pass by ref?
69. What are the advantages of cross-coverage?
70. What is the difference between associative and dynamic array?
71. What is the type of SystemVerilog assertions?
72. What is the difference between *display*, *strobe*, *\$monitor*?
73. Can we write SystemVerilog assertions in class?
74. What is an argument pass by value and pass by reference?
75. what is the difference between a bit and logic data type in system verilog?
76. what is the difference between \$display, \$write, \$monitor, and \$strobe in systemverilog?
77. what is the difference between a packed array and an unpacked array in system verilog with example?
78. suppose a dynamic array of integers is initialized to values as shown below. write a code to find all elements greater than 5 in the array using array locator method “find”?
79. how can we use system verilog constraints to generate a dynamic array with random but unique values with and without unique constraint?
80. explain the concept of a “ref” and “const ref” argument in systemverilog function or task with example?
81. difference between “forever loop” and “for loop” in systemverilog?
82. what is the difference between “case”, “casex” and “casez” in systemverilog?
83. how regions inside a system verilog simulation semantics works? explain with examples
84. what is a modport in an system verilog interface?
85. which of the logical equality operators “==” or “===” are used in case expression conditions for case, casex, and casez?
86. what is the difference between byte and logic[7:0] variable in system verilog?
87. what is the difference between input #1 step sig_1; and input #1ns sig_1; ways of specifying skews in a clocking block?
88. given a dynamic array of size 50, how can the array be re-sized to hold 100 elements while

- the lower 50 elements are preserved as original in system verilog?
89. what are `pre_randomize()` and `post_randomize()` in system verilog with example?
 90. what is the difference between a union and struct in systemverilog with example?
 91. explain unique constraint in system verilog with example?
 92. which of the array types are good to model really large arrays or a huge memory of 16/32/64kb?
 93. difference between private, public, and protected data members of a systemverilog class.
 94. what is the difference between `new()` and `new[]` in systemverilog?
 95. what is difference between bounded and unbounded mailboxes in system verilog? explain with examples
 96. explain interfaces in system verilog with example
 97. what is the difference between a logic, reg and wire in system verilog?
 98. explain system tasks and functions in system verilog? give some example of system tasks and functions
 99. explain the concept of forward declaration of a class in system verilog with example.
 100. what metrics would you use to track the progress of verification project?
 101. in system verilog how can we merge two events? explain with example
 102. explain clocking block with example and what are the advantages of using clocking blocks inside system verilog interface?
 103. identify how many parallel threads does this given code executes?
 104. how can we disable or enable constraints selectively in a system verilog class?
 105. given a class with following constraints, how can you generate a packet object with address value greater than 500?
 106. write a system verilog constraint to generate a random value such that it always has 15 bits as 1 and no two bits next to each other should be 1
 107. what is gate level simulation and why is it important in signing off any design?
 108. write system verilog constraints to generate elements of a dynamic array such that each element of the array is less than 50 and the array size is less than 50.
 109. write system verilog constraints to create a random array such that array size is between 50 and 100 and the values of the array are in descending order?
 110. hard constraints vs soft constraints with examples
 111. is the given systemverilog constraint correct? if yes what will be the output and if no what's wrong with it. please explain
 112. how to override a constraint defined in the base/parent class in the derived/child class?
 113. when can you say that verification is complete?

10 HDL Theory questions

- HDL Theory questions
 - Verilog
 - SystemVerilog
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11 Verilog

1. Write a Verilog code RS NAND latch

2. Write a Verilog code D-flip flop in the behavioural model (async reset with active low)
 3. Write a Verilog code for a 4-bit counter
 4. Write a Verilog code for swapping two numbers (with & without the third variable)
 5. Write a Verilog for 4:1 MUX in gate level, dataflow & behavioural modelling
 6. Write the Verilog code for full adder using different styles
 7. Write a Verilog code for full subtractor
 8. Write a Verilog code for a 4 to 2 priority encoder using dataflow modelling
 9. Write a Verilog code for the 011 sequence (both Mealy & Moore- overlap & non-overlap)
 10. Write a Verilog code for the Mod-35 counter
 11. Write a Verilog code for the frequency divider by 3 circuit
 12. Write a Verilog code for a 4-bit up-down counter
 13. Write a Verilog code for the 1MHz clock generator
 14. Write Verilog Code for JK FlipFlop
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12 SystemVerilog